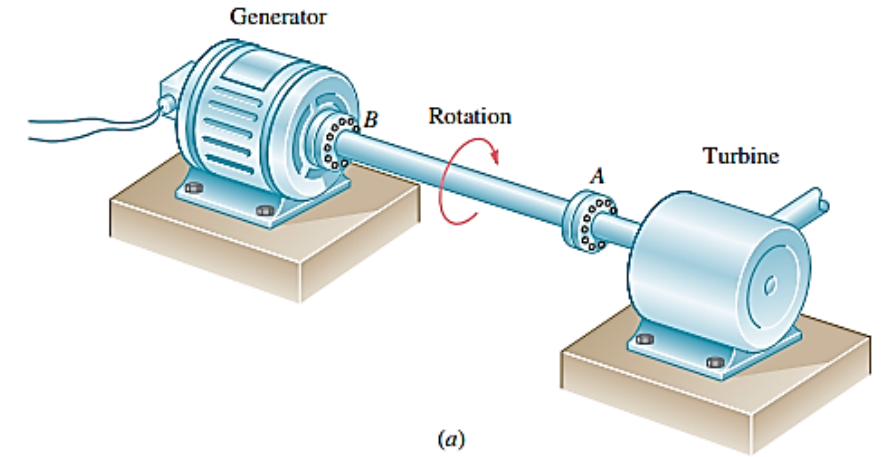


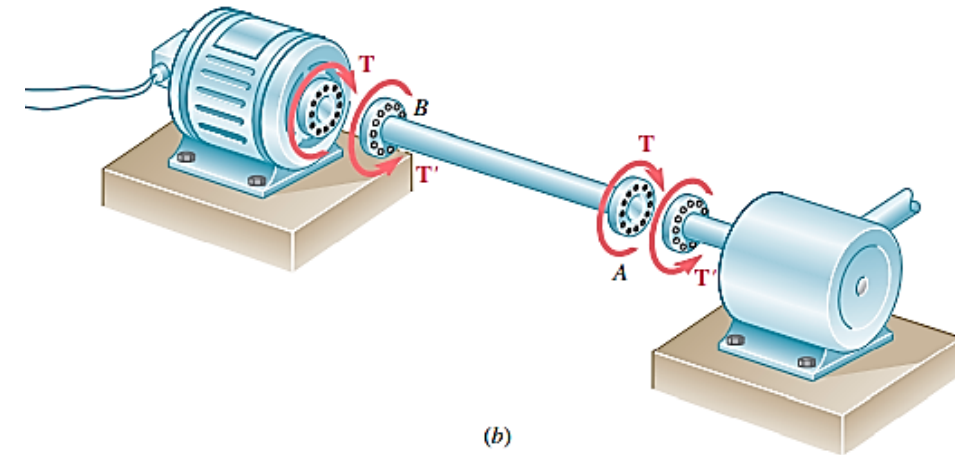


Strength of Materials

Torsion



(a)



(b)

Inst. : Dawit M.



Chapter 4: Torsion

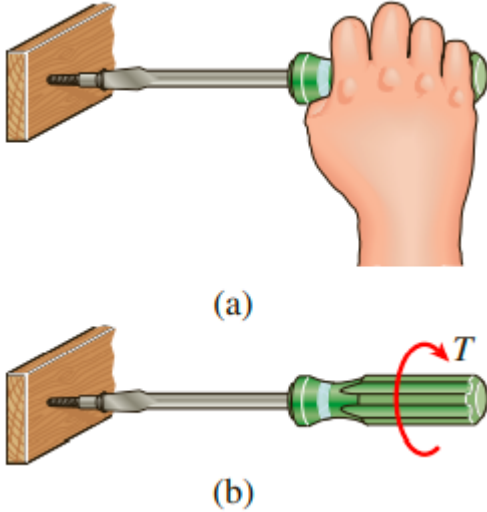
A large crankshaft is needed to turn the propeller shaft to power large ships. The propeller shaft develops torsional stresses and twisting deformations during rotation.

Contents

- Introduction
- Torsional Loads on Circular Shafts
- Net Torque Due to Internal Stresses
- Axial Shear Components
- Shaft Deformations Shearing Strain
- Stresses in Elastic Range
- Normal Stresses
- Torsional Failure Modes
- Sample Problems
- Angle of Twist in Elastic Range

Torsion

Torsion of a screwdriver due to a torque T applied to the handle



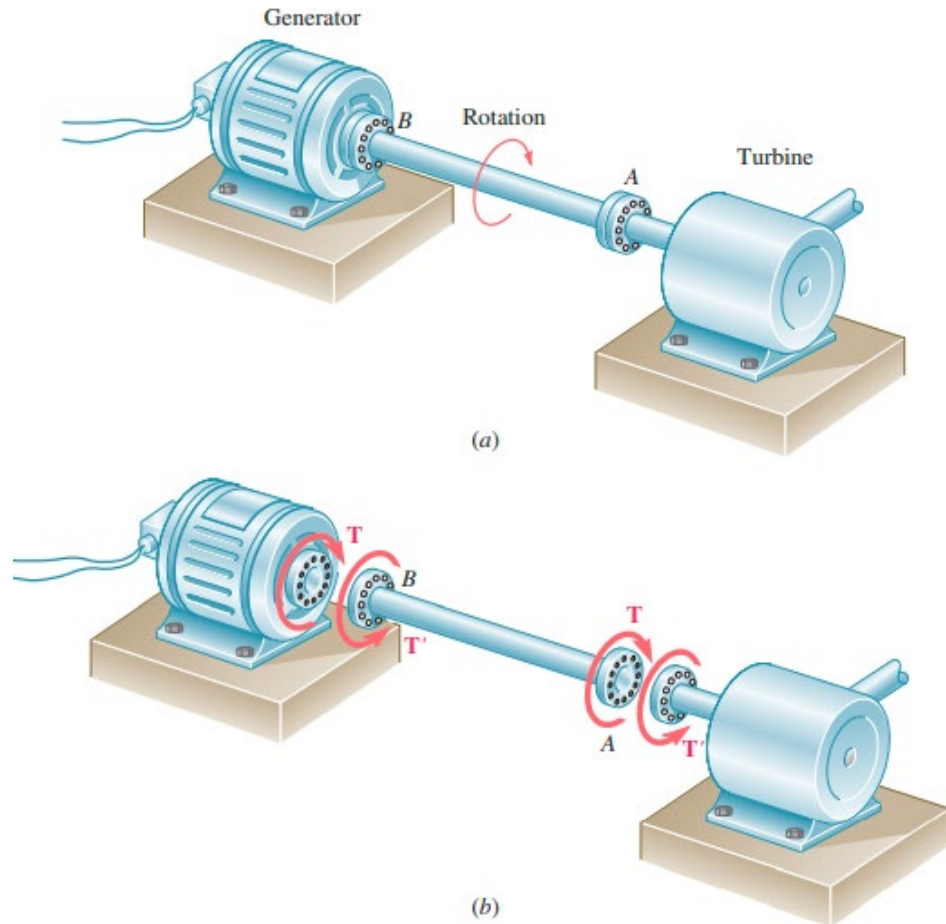
- **Introduction:**

- In this chapter, structural members and machine parts that are in torsion will be analyzed, where the stresses and strains in members of circular cross section are subjected to twisting couples, or torques, T and T'
- Members in torsion are encountered in many engineering applications. The most common application is provided by transmission shafts, which are used to transmit power from one point to another



In this automotive power train, the shaft transmits power from the engine to the rear wheels.

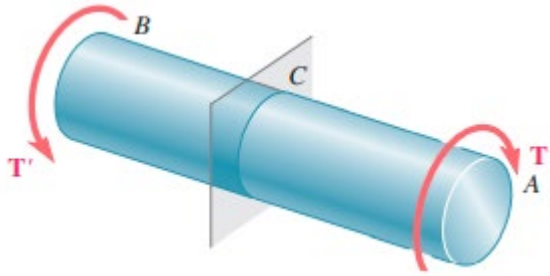
Torsional Loads on Circular Shafts



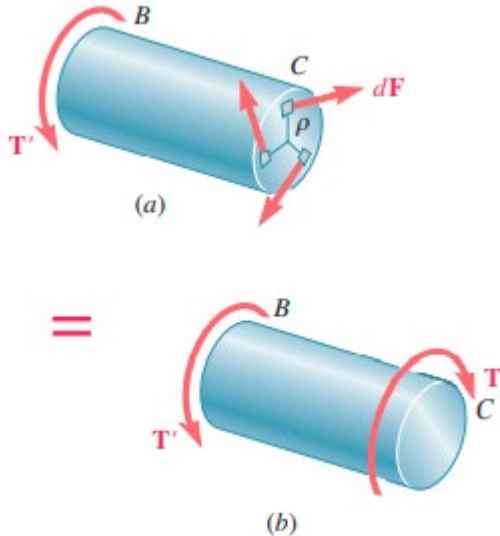
- Interested in stresses and strains of circular shafts subjected to twisting couples or *torques*
- Turbine exerts torque T on the shaft
- Shaft transmits the torque to the generator
- Generator creates an equal and opposite torque T'

- (a) A generator receives power at a constant number of revolutions per minute from a turbine through shaft AB.
- (b) Free-body diagram of shaft AB along with the driving and reacting torques on the generator and turbine, respectively

Net Torque Due to Internal Stresses



Shaft subject to torques, with a section plane at C.



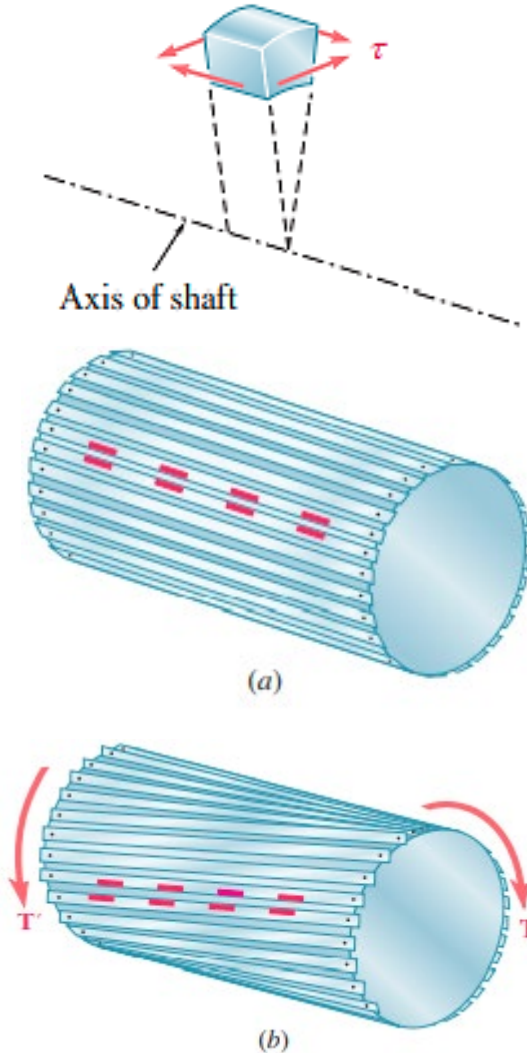
a) Free-body diagram of section BC with torque at C represented by the contributions of small elements of area carrying forces dF at a radius ρ from the section center. (b) Free-body diagram of section BC having all the small area elements summed, resulting in torque T .

- Net of the internal shearing stresses is an internal torque, equal and opposite to the applied torque,

$$T = \int \rho \, dF = \int \rho (\tau \, dA)$$

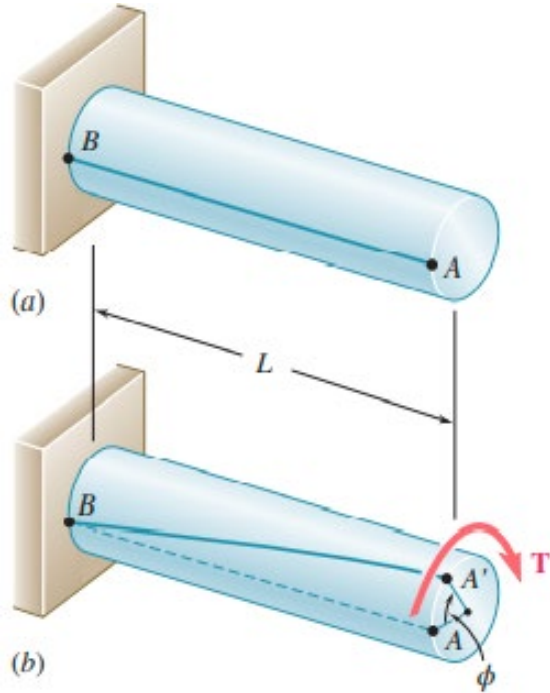
- Although the net torque due to the shearing stresses is known, the distribution of the stresses is not
- Distribution of shearing stresses is statically indeterminate – must consider shaft deformations
- Unlike the normal stress due to axial loads, the distribution of shearing stresses due to torsional loads can not be assumed uniform.

Axial Shear Components



- Torque applied to shaft produces shearing stresses on the faces perpendicular to the axis.
- Conditions of equilibrium require the existence of equal stresses on the faces of the two planes containing the axis of the shaft
- The existence of the axial shear components is demonstrated by considering a shaft made up of axial slats.
- The slats slide with respect to each other when equal and opposite torques are applied to the ends of the shaft.

Shaft Deformations

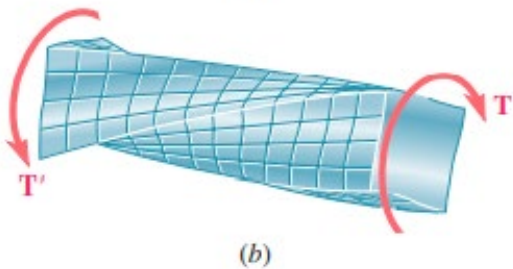
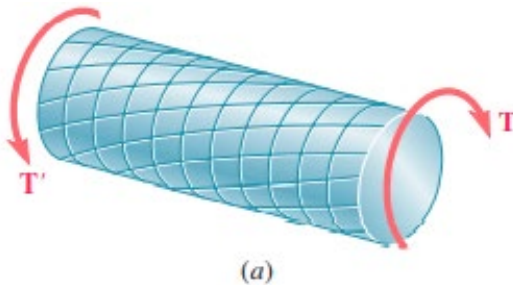


- From observation, the angle of twist of the shaft is proportional to the applied torque and to the shaft length.

$$\phi \propto T$$

$$\phi \propto L$$

- When subjected to torsion, every cross-section of a circular shaft remains plane and undistorted.
- Cross-sections for hollow and solid circular shafts remain plane and undistorted because a circular shaft is axisymmetric.
- Cross-sections of noncircular (non-axisymmetric) shafts are distorted when subjected to torsion.



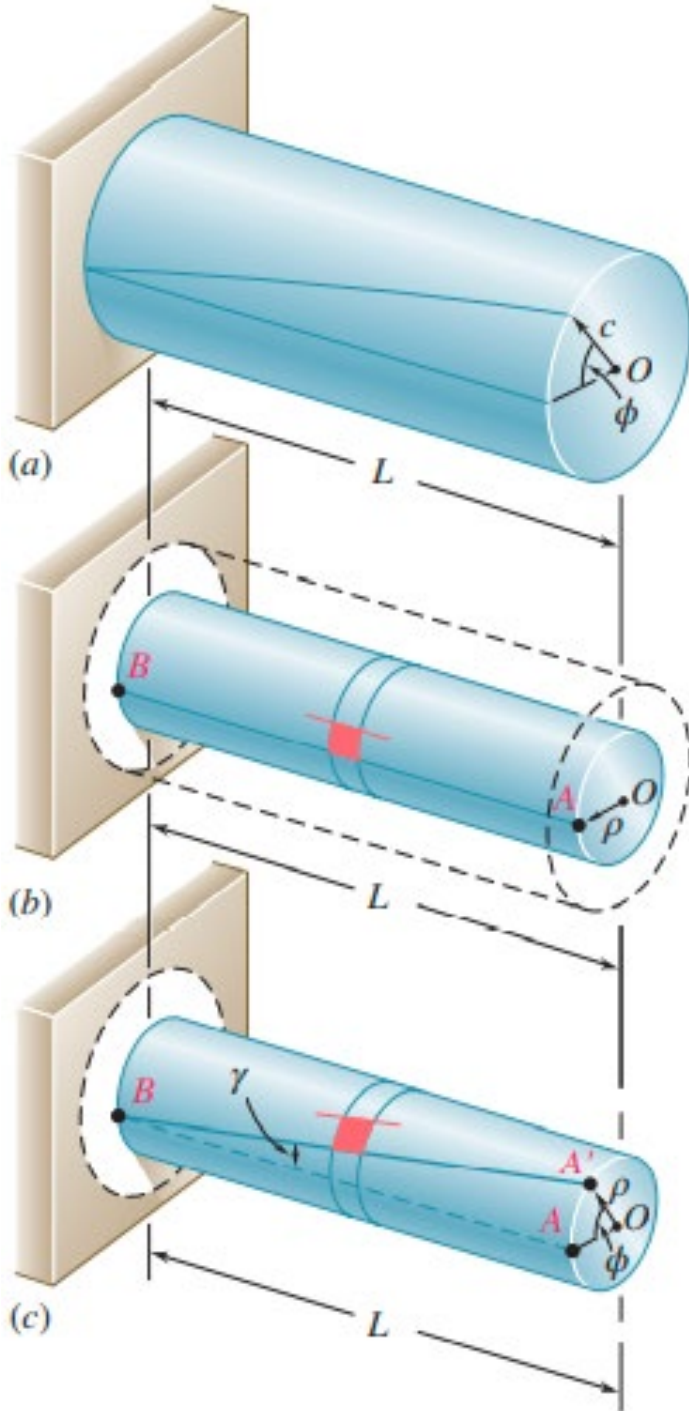
Shearing Strain

- Consider an interior section of the shaft. As a torsional load is applied, an element on the interior cylinder deforms into a rhombus.
- Since the ends of the element remain planar, the shear strain is equal to angle of twist.
- It follows that

$$L\gamma = \rho\phi \quad \text{or} \quad \gamma = \frac{\rho\phi}{L}$$

- Shear strain is proportional to twist and radius

$$\gamma_{\max} = \frac{c\phi}{L} \quad \text{and} \quad \gamma = \frac{\rho}{c}\gamma_{\max}$$



Stresses in Elastic Range

- Multiplying the previous equation by the shear modulus,

$$G\gamma = \frac{\rho}{c} G\gamma_{\max}$$

From Hooke's Law, $\tau = G\gamma$, so

$$\tau = \frac{\rho}{c} \tau_{\max}$$

The shearing stress varies linearly with the radial position in the section.

- Recall that the sum of the moments from the internal stress distribution is equal to the torque on the shaft at the section,

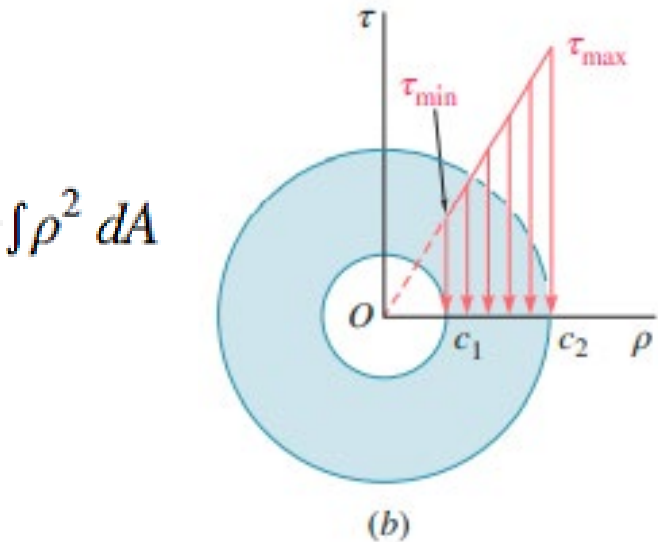
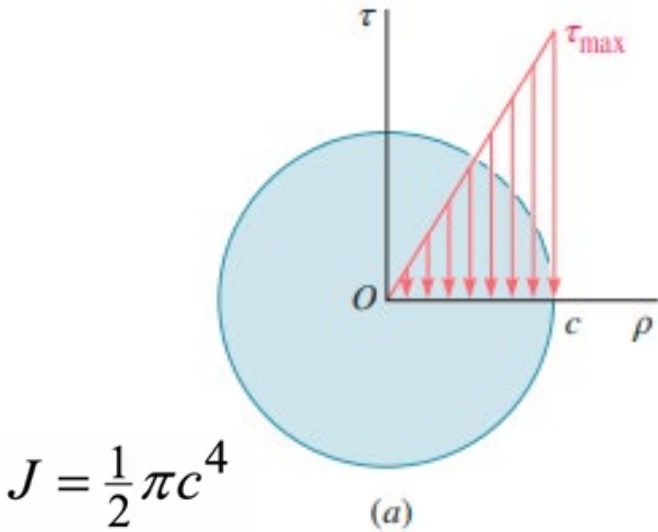
$$T = \int \rho \tau dA = \frac{\tau_{\max}}{c} \int \rho^2 dA = \frac{\tau_{\max} J}{c}$$

- The results are known as the elastic torsion formulas,

$$\tau_{\max} = \frac{Tc}{J}$$

and

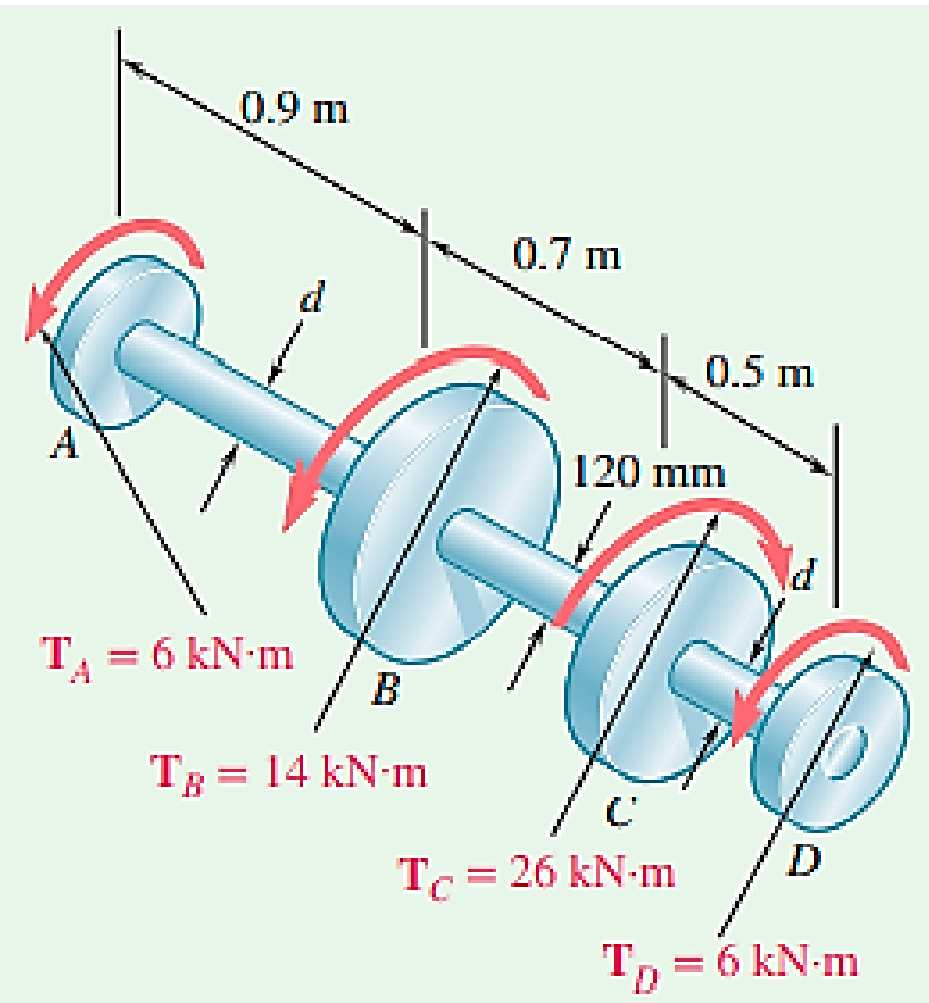
$$\tau = \frac{T\rho}{J}$$



$$J = \frac{1}{2} \pi (c_2^4 - c_1^4)$$

Example

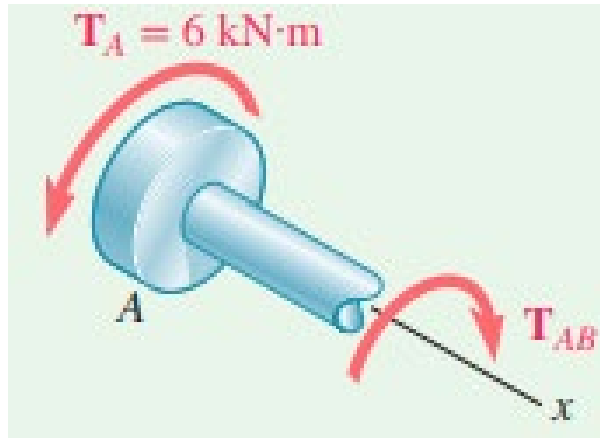
- Shaft BC is hollow with inner and outer diameters of 90 mm and 120 mm, respectively. Shafts AB and CD are solid of diameter d .
- For the loading shown, determine (a) the minimum and maximum shearing stress in shaft BC , (b) the required diameter d of shafts AB and CD if the allowable shearing stress in these shafts is 65 MPa.



SOLUTION:

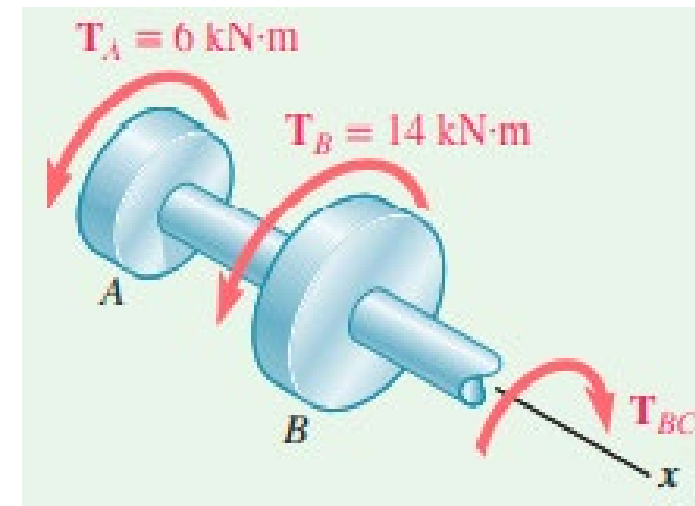
- Cut sections through shafts AB and BC and perform static equilibrium analysis to find torque loadings
- Apply elastic torsion formulas to find minimum and maximum stress on shaft BC
- Given allowable shearing stress and applied torque, invert the elastic torsion formula to find the required diameter

- Cut sections through shafts AB and BC and perform static equilibrium analysis to find torque loadings



$$\sum M_x = 0 = (6 \text{ kN} \cdot \text{m}) - T_{AB}$$

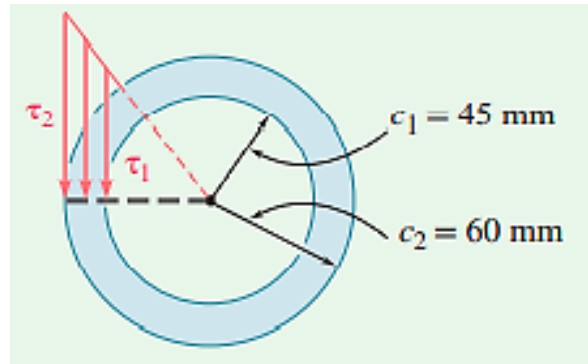
$$T_{AB} = 6 \text{ kN} \cdot \text{m} = T_{CD}$$



$$\sum M_x = 0 = (6 \text{ kN} \cdot \text{m}) - T_{AB}$$

$$T_{AB} = 6 \text{ kN} \cdot \text{m} = T_{CD}$$

- Apply elastic torsion formulas to find minimum and maximum stress on shaft *BC*



$$J = \frac{\pi}{2}(c_2^4 - c_1^4) = \frac{\pi}{2}[(0.060)^4 - (0.045)^4]$$

$$= 13.92 \times 10^{-6} \text{ m}^4$$

$$\tau_{\max} = \tau_2 = \frac{T_{BC}c_2}{J} = \frac{(20 \text{ kN} \cdot \text{m})(0.060 \text{ m})}{13.92 \times 10^{-6} \text{ m}^4}$$

$$= 86.2 \text{ MPa}$$

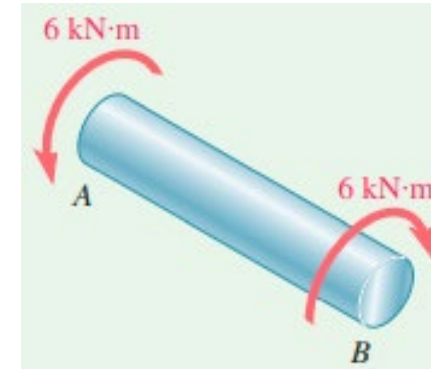
$$\frac{\tau_{\min}}{\tau_{\max}} = \frac{c_1}{c_2} \quad \frac{\tau_{\min}}{86.2 \text{ MPa}} = \frac{45 \text{ mm}}{60 \text{ mm}}$$

$$\tau_{\min} = 64.7 \text{ MPa}$$

$$\tau_{\max} = 86.2 \text{ MPa}$$

$$\tau_{\min} = 64.7 \text{ MPa}$$

- Given allowable shearing stress and applied torque, invert the elastic torsion formula to find the required diameter



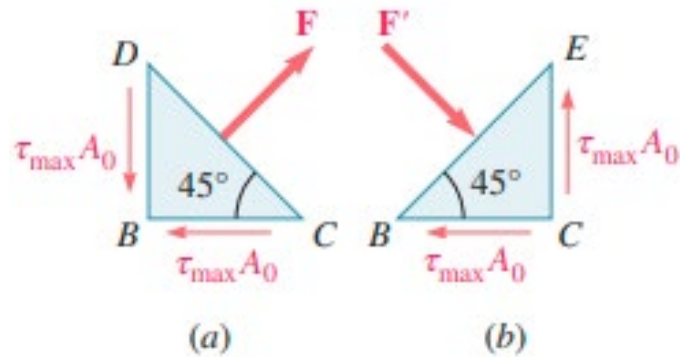
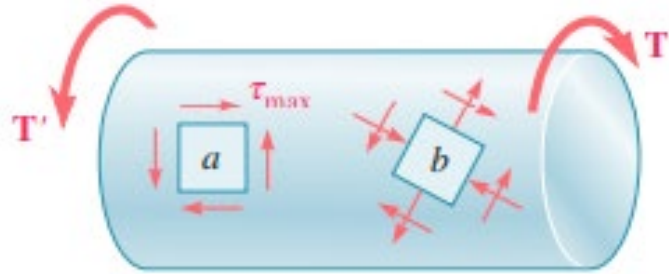
$$\tau_{\max} = \frac{Tc}{J} = \frac{Tc}{\frac{\pi}{2}c^4}$$

$$c = 38.9 \times 10^{-3} \text{ m}$$

$$65 \text{ MPa} = \frac{6 \text{ kN} \cdot \text{m}}{\frac{\pi}{2}c^3}$$

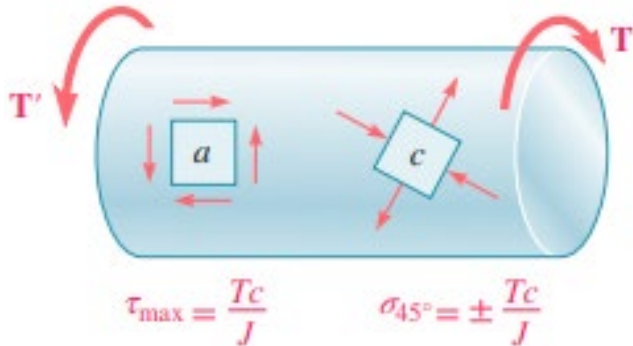
$$d = 2c = 77.8 \text{ mm}$$

Normal Stresses



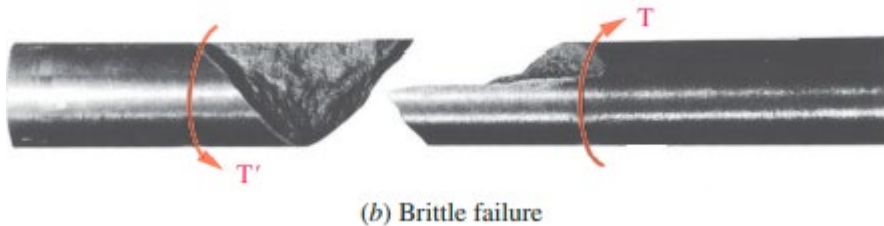
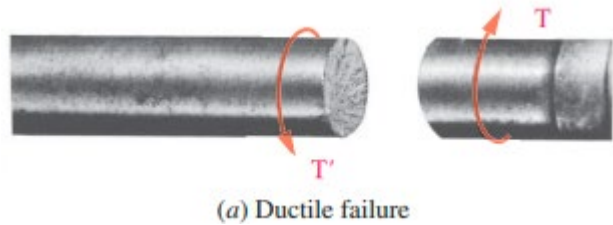
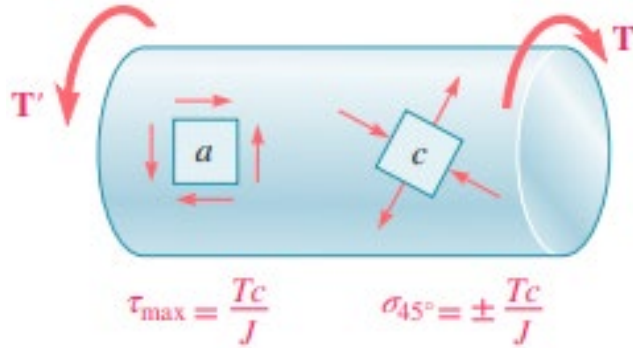
$$F = 2(\tau_{\max} A_0) \cos 45 = \tau_{\max} A_0 \sqrt{2}$$

$$\sigma_{45^\circ} = \frac{F}{A} = \frac{\tau_{\max} A_0 \sqrt{2}}{A_0 \sqrt{2}} = \tau_{\max}$$



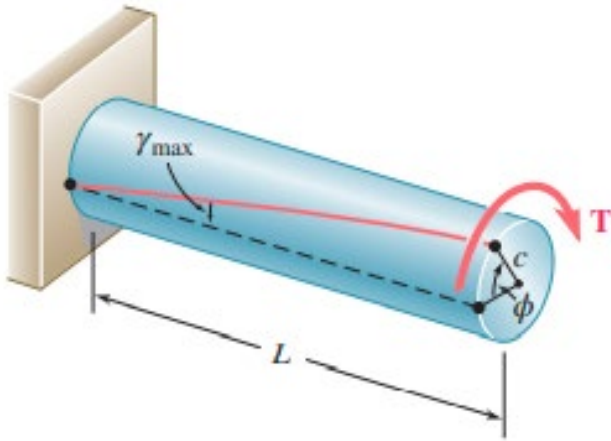
- Elements with faces parallel and perpendicular to the shaft axis are subjected to shear stresses only. Normal stresses, shearing stresses or a combination of both may be found for other orientations.
 - Consider an element at 45° to the shaft axis,
- $$F = 2(\tau_{\max} A_0) \cos 45 = \tau_{\max} A_0 \sqrt{2}$$
- $$\sigma_{45^\circ} = \frac{F}{A} = \frac{\tau_{\max} A_0 \sqrt{2}}{A_0 \sqrt{2}} = \tau_{\max}$$
- Element a is in pure shear.
 - Element c is subjected to a tensile stress on two faces and compressive stress on the other two.
 - Note that all stresses for elements a and c have the same magnitude

Torsional Failure Modes



- Ductile materials generally fail in shear. Brittle materials are weaker in tension than shear.
- When subjected to torsion, a ductile specimen breaks along a plane of maximum shear, i.e., a plane perpendicular to the shaft axis.
- When subjected to torsion, a brittle specimen breaks along planes perpendicular to the direction in which tension is a maximum, i.e., along surfaces at 45° to the shaft axis.

Angle of Twist in Elastic Range



- Recall that the angle of twist and maximum shearing strain are related,

$$\gamma_{\max} = \frac{c\phi}{L}$$

- In the elastic range, the shearing strain and shear are related by Hooke's Law,

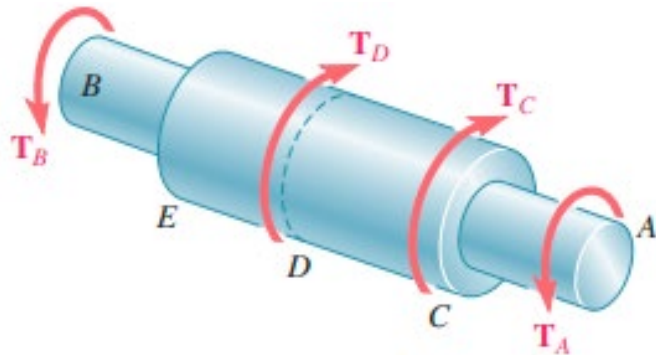
$$\gamma_{\max} = \frac{\tau_{\max}}{G} = \frac{Tc}{JG}$$

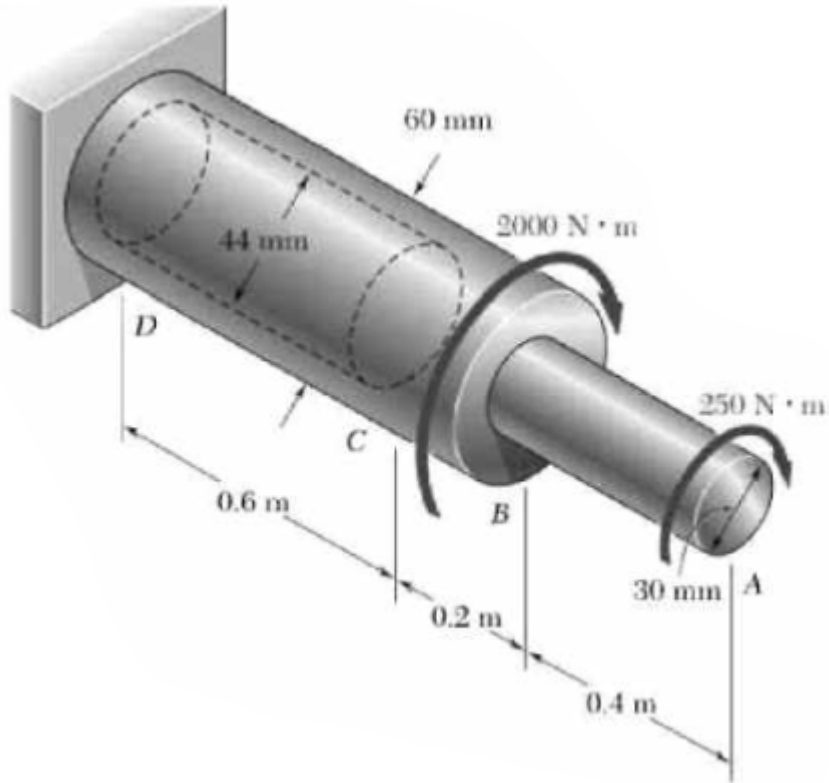
- Equating the expressions for shearing strain and solving for the angle of twist,

$$\phi = \frac{TL}{JG}$$

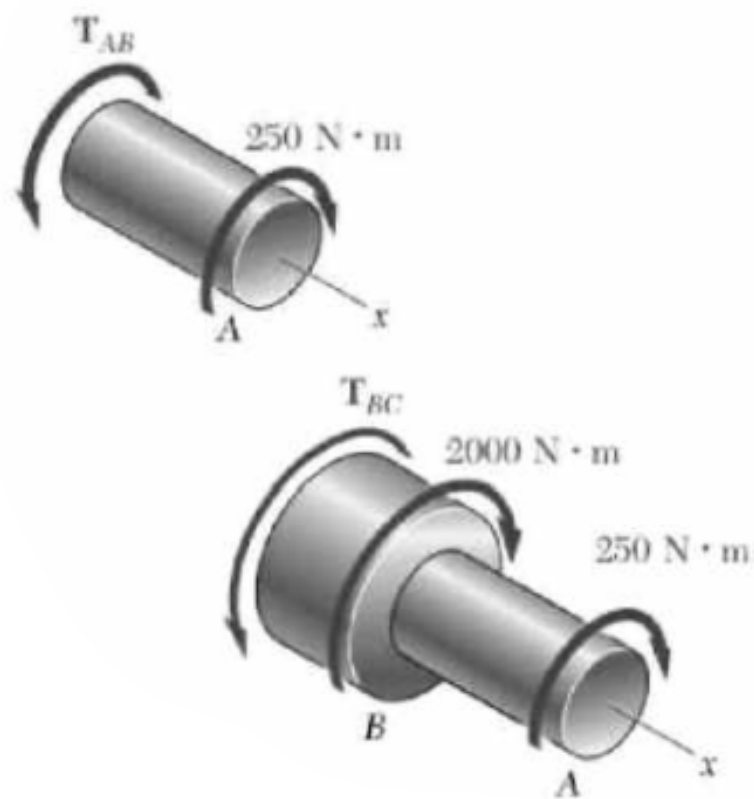
- If the torsional loading or shaft cross-section changes along the length, the angle of rotation is found as the sum of segment rotations

$$\phi = \sum_i \frac{T_i L_i}{J_i G_i}$$





- The horizontal shaft AD is attached to a fixed base at P and is subjected to the torques shown. A 44-mm-diameter hole has been drilled into portion CD of the shaft. Knowing that the entire shaft is made of steel for which $G = 77 \text{ GPa}$.
- Determine the angle of twist at end A



SOLUTION

Since the shaft consists of three portions AB , BC , and CD , each of uniform cross section and each with a constant internal torque, Eq. (3.17) may be used.

Statics. Passing a section through the shaft between A and B and using the free body shown, we find

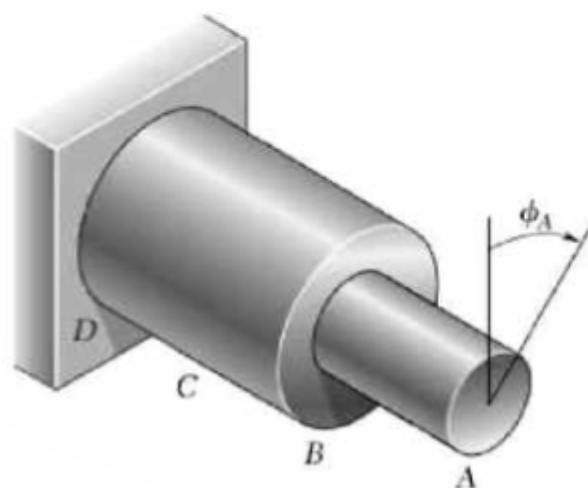
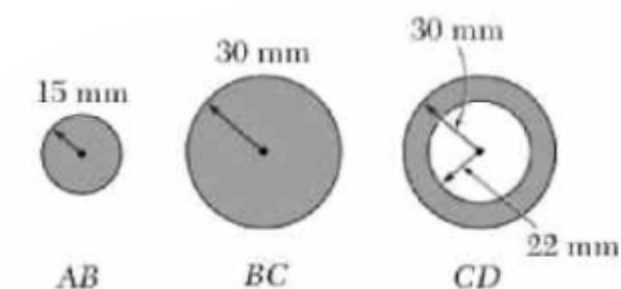
$$\Sigma M_x = 0: \quad (250 \text{ N} \cdot \text{m}) - T_{AB} = 0 \quad T_{AB} = 250 \text{ N} \cdot \text{m}$$

Passing now a section between B and C , we have

$$\Sigma M_x = 0: \quad (250 \text{ N} \cdot \text{m}) + (2000 \text{ N} \cdot \text{m}) - T_{BC} = 0 \quad T_{BC} = 2250 \text{ N} \cdot \text{m}$$

Since no torque is applied at C ,

$$T_{CD} = T_{BC} = 2250 \text{ N} \cdot \text{m}$$



Polar Moments of Inertia

$$J_{AB} = \frac{\pi}{2} c^4 = \frac{\pi}{2} (0.015 \text{ m})^4 = 0.0795 \times 10^{-6} \text{ m}^4$$

$$J_{BC} = \frac{\pi}{2} c^4 = \frac{\pi}{2} (0.030 \text{ m})^4 = 1.272 \times 10^{-6} \text{ m}^4$$

$$J_{CD} = \frac{\pi}{2} (c_2^4 - c_1^4) = \frac{\pi}{2} [(0.030 \text{ m})^4 - (0.022 \text{ m})^4] = 0.904 \times 10^{-6} \text{ m}^4$$

Angle of Twist. Using Eq. (3.17) and recalling that $G = 77 \text{ GPa}$ for the entire shaft, we have

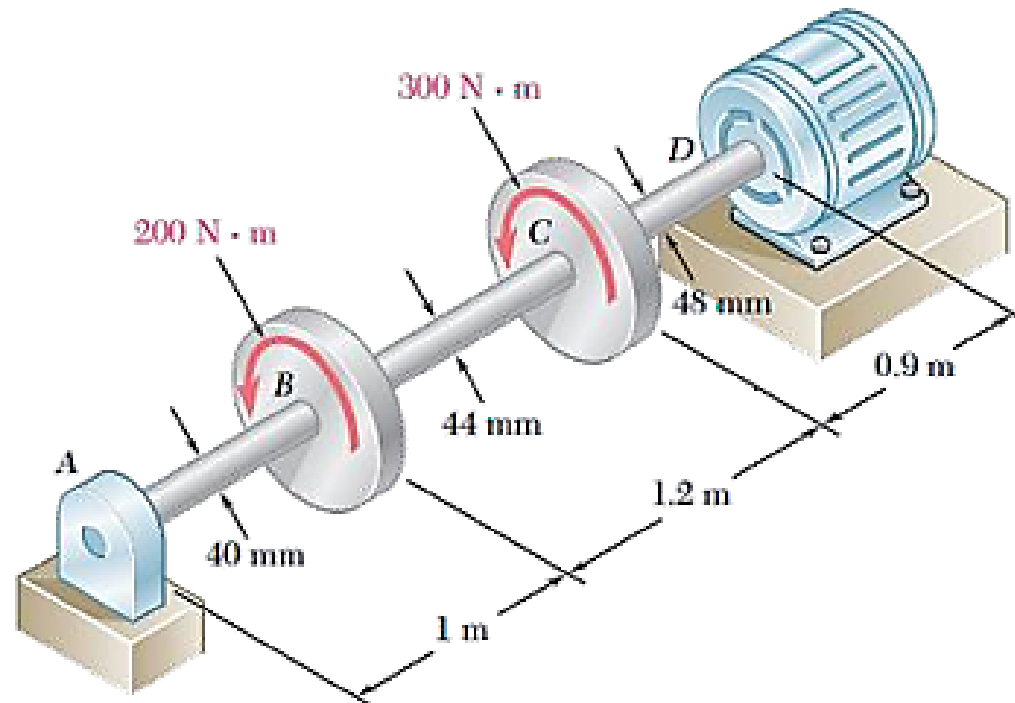
$$\begin{aligned} \phi_A &= \sum_i \frac{T_i L_i}{J_i G} = \frac{1}{G} \left(\frac{T_{AB} L_{AB}}{J_{AB}} + \frac{T_{BC} L_{BC}}{J_{BC}} + \frac{T_{CD} L_{CD}}{J_{CD}} \right) \\ \phi_A &= \frac{1}{77 \text{ GPa}} \left[\frac{(250 \text{ N} \cdot \text{m})(0.4 \text{ m})}{0.0795 \times 10^{-6} \text{ m}^4} + \frac{(2250)(0.2)}{1.272 \times 10^{-6}} + \frac{(2250)(0.6)}{0.904 \times 10^{-6}} \right] \\ &= 0.01634 + 0.00459 + 0.01939 = 0.0403 \text{ rad} \end{aligned}$$

$$\phi_A = (0.0403 \text{ rad}) \frac{360^\circ}{2\pi \text{ rad}}$$

$$\phi_A = 2.31^\circ \quad \blacktriangleleft$$

Ex. HW

- The electric motor exerts a 500 N.m-torque on the aluminum shaft *ABCD* when it is rotating at a constant speed. Knowing that $G = 27$ GPa and that the torques exerted on pulleys *B* and *C* are as shown, determine the angle of twist between (a) *B* and *C*, (b) *B* and *D*.



Thank you

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